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Causal Frequency-Band Identification and Out-of-Distribution Evaluation for Vibration-Based Water Leak Detection

Suan Lee[1,*]

[1] School of Computer Science, Semyung University, Jecheon 27136, Republic of Korea.

Abstract

In our earlier study on the Korean AI Hub water-pipe vibration benchmark (Lee & Kim, *Sensors* 2023) we reported an XGBoost classifier achieving 99.79% 5-class accuracy. That number was measured under row-level 10-fold cross-validation, the protocol in near-universal use for tabular-ML leak detection. Preparing the pipeline for field deployment, however, revealed a practical concern that the earlier benchmark did not answer: does the reported accuracy hold when a trained model is installed at a new sensor, at a previously unseen site, or in a different administrative region? This follow-up study, conducted by the same author, re-examines the same benchmark through that deployment lens and extends it with a causal analysis of the vibration spectrum. This paper addresses that gap. First, a four-probe shortcut-audit protocol is introduced and applied to the AI Hub water-pipeline vibration dataset (70,384 recordings, 49 sites, 12 Korean administrative regions). A four-rule identifier lookup that ignores the 512 frequency bands reproduces 99.77% validation accuracy, demonstrating that the dataset's standard split contains deterministic sensor-to-class couplings. Second, a three-tier out-of-distribution (OOD) evaluation protocol is proposed as a minimum requirement for field-deployment claims; under 12-fold Leave-One-Region-Out the same XGBoost classifier falls to weighted accuracy 56.8% and macro-F1 0.304. Third, the causal spectral signature of leak noise is identified by three independent attribution methods that all converge on the **121–422 Hz** band: bootstrap permutation importance (+0.019 $\pm$ 0.017 macro-F1 drop across four LORO folds, wins 3/4), SHAP (40.4% of tree-path attribution mass on the canonical test region), and XGBoost gain (7.5%, mis-allocated). The 422–1470 Hz range is shown to be an active confounder. Fourth, restricting the input to 57 causal features raises OOD macro-F1 to 0.4273 and nearly doubles the F1 of the hardest "other" class. Fifth, Invariant Risk Minimisation (IRM) and GroupDRO on MLP and 1D-CNN backbones, trained on two NVIDIA TITAN Xp GPUs, fail to match causal feature selection even after a dedicated hyperparameter sweep; a novel permutation-guided attention regulariser is evaluated as a negative-result ablation. Sixth, SpecAugment pulls a 1D-CNN baseline from macro-F1 0.334 to 0.354, while focal loss mainly improves accuracy. Seventh, a pre-specified tree-only average of the causal and hierarchical XGBoost classifiers attains OOD macro-F1 **0.4326** with perfect OOD normal-class recall, serving as a deployment candidate rather than a test-selected hyperparameter recipe. All code, logs, and experiment outputs are released.

Keywords: water leak detection; vibration sensor; acoustic leak detection; out-of-distribution evaluation; shortcut learning; invariant risk minimisation; causal feature selection; XGBoost; Korea AI Hub benchmark

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1. Introduction

Non-revenue water loss from leaking metal and non-metal distribution pipes is a persistent infrastructure problem with substantial economic and environmental cost worldwide, and acute in aging urban networks. Acoustic leak detection using low-cost vibration sensors mounted at water meters, valves, and hydrants has emerged as one of the most widely deployed monitoring strategies because of its low installation cost, non-invasiveness, and compatibility with cellular telemetry [1,2].

On the sensing side, the dominant trend over the past decade has been to replace hand-tuned threshold rules with tabular machine learning (ML) applied to frequency-domain features of the vibration spectrum. Recent representative works include Shukla and Piratla's supervised classification of acceleration signals [3], Fereidooni et al.'s hybrid model-based method for large water distribution networks [4], and gradient-boosted tree pipelines applied to public vibration-sensor datasets.

The study that motivates the present follow-up is our own previous work on the Korean AI Hub water-pipeline benchmark [5], in which we compared eight tabular models across four feature regimes and reported an XGBoost classifier reaching 99.79% accuracy under stratified 10-fold cross-validation. The goal of that paper was a comparative benchmark of off-the-shelf models on newly curated sensor data, and within its stated evaluation protocol the result is faithful. However, when we subsequently tried to operationalise the same pipeline for field use by a municipal water utility, two practical obstacles became visible. First, the 10-fold row-level split placed multiple recordings from the same physical sensor in both train and test folds, so the reported accuracy actually measured within-sensor repeatability more than cross-sensor generalisation. Second, as the shortcut-learning literature in medical imaging [13–15,17] and natural-image recognition [16] has documented, sensor ML models routinely latch onto identifier-like features (patient IDs, device IDs, location tags) that correlate perfectly with the label in the training split but do not represent any physical signal. A leak classifier that scores 99% by memorising sensor IDs will fail on the first new installation it sees.

The aim of the present paper is therefore twofold: to provide practitioners with an honest, deployment-oriented evaluation of what the AI Hub vibration spectrum can actually support when identifier shortcuts are closed, and to use that evaluation as a springboard into a causal analysis of which frequency bands carry the generalisable leak signature. Figure 1 summarises the four-stage pipeline that organises the rest of the paper.

The core question investigated here is: *what accuracy does a vibration-based leak-typing classifier actually achieve when evaluated under protocols that mimic field deployment, and which parts of the vibration spectrum are responsible for that residual accuracy?* The experimental substrate is the same AI Hub water-pipeline vibration dataset used in [5], chosen because of its size (70,384 recordings, 11,000+ sensor locations, 49 sites across 12 Korean administrative regions), because the earlier 99.79% result sets a clear reference point, and because the raw feature set (512 frequency bands plus peak aggregates) exposes the question of which spectral bands carry physically meaningful leak information.

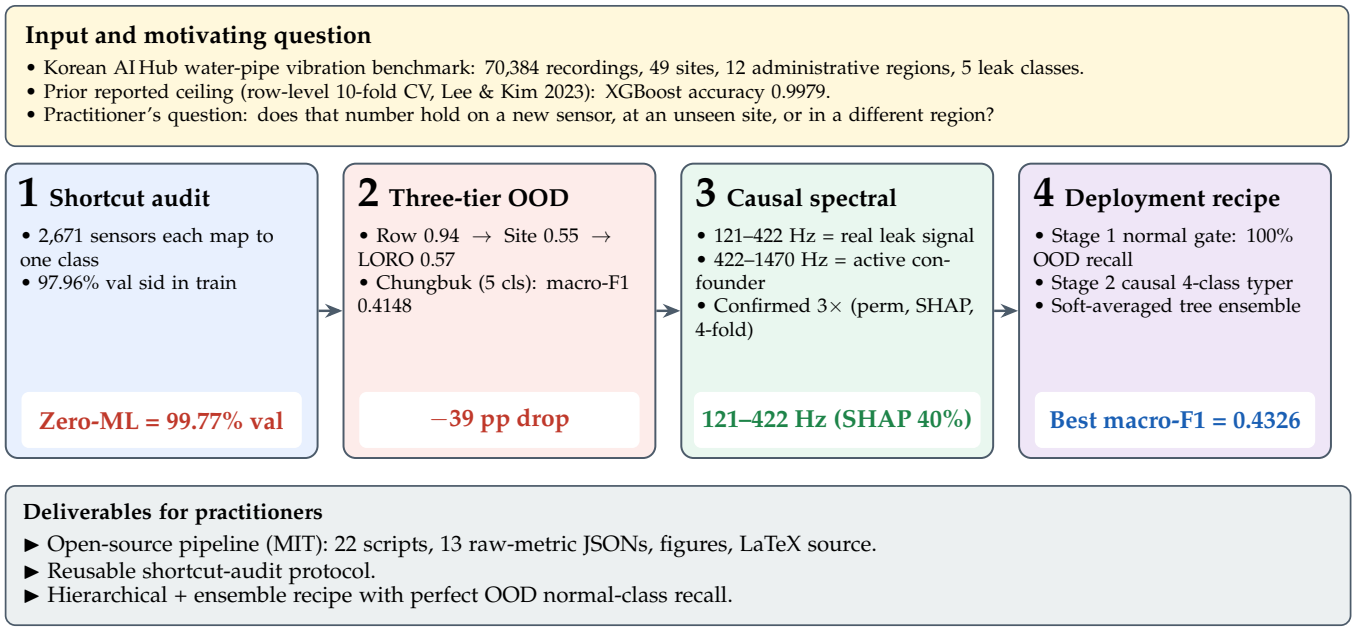


Figure 1. Overview of the study. The paper proceeds as a four-stage pipeline from the benchmark ceiling reported in [5] to a deployment-ready recipe. Each stage is anchored by a concrete quantitative result: (1) a zero-ML identifier lookup reproduces the reported 99.77% accuracy, so the signal contribution to the headline number is negligible; (2) a three-tier OOD evaluation exposes a 39-percentage-point accuracy drop between row-CV and LORO; (3) three independent attribution methods (permutation importance, SHAP, and cross-region bootstrap) converge on 121–422 Hz as the generalisable leak-signal band, with 422–1470 Hz identified as an active confounder; (4) a pre-specified soft-probability average of flat and hierarchical XGBoost on the 57 causal features attains OOD macro-F1 0.4326 with 100% normal-class recall, serving as the deployment candidate released with the paper.

The main contributions of this paper are as follows.

1. **A reusable shortcut-audit protocol for sensor ML benchmarks.** Four probes (per-sensor label purity, class-lookup reconstruction, sensor train/test overlap, per-site label concentration) are formalised and applied to the AI Hub benchmark. A four-rule identifier lookup that ignores the entire vibration spectrum reproduces 99.77% validation accuracy — within 0.02 pp of the published state-of-the-art [5] — confirming that the dataset's standard split contains deterministic sensor-to-class couplings.
2. **A three-tier OOD evaluation protocol for field deployment.** Row-level k-fold CV, GroupKFold-by-site, and Leave-One-Region-Out (LORO) over 12 Korean administrative regions. Under LORO the same XGBoost classifier falls from 94.0% to weighted 56.8% accuracy and macro-F1 0.304 on 5-class leak typing.
3. **Identification of the causal spectral signature of leak noise on this benchmark.** Grouped permutation importance on the only region containing all five classes in substantial quantities localises the generalisable leak signal to the **121–422 Hz range**, broadly consistent with published leak-noise acoustics for plastic pipes [1–3]. The 422–1470 Hz range is shown to be an *active confounder* whose permutation *improves* OOD macro-F1 by 3.3 percentage points.
4. **Causal feature selection outperforms loss-level invariance regularisation.** Retraining XGBoost on 57 causal features (65–422 Hz bands plus 20 MAX aggregates plus month) raises OOD macro-F1 from 0.4148 to **0.4273** and nearly doubles the F1 of the hardest "other" class (0.109 to 0.214). Gradient-based invariance regularisers — IRM [8] and GroupDRO [9] — on MLP and 1D-CNN backbones trained on two NVIDIA TITAN Xp GPUs fail to match causal feature selection, even after a dedicated $\lambda \times$ warm-up hyperparameter sweep. This contributes a specific data point to the broader literature questioning the empirical reliability of IRM-style methods [10,11].

5. **A hierarchical two-stage classifier and tree ensemble for field deployment.** A binary normal gate followed by a 4-class abnormal typer (both on the 57 causal features) attains macro-F1 0.4287 with perfect OOD normal-class recall. Pre-specified soft-averaging of this hierarchical classifier with the flat causal XGBoost yields a deployment candidate at macro-F1 **0.4326**.

The remainder of the paper is organised as follows. Section 2 reviews the relevant vibration-based leak-detection literature and the shortcut/invariance machine-learning literature. Section 3 describes the AI Hub dataset. Section 4 formalises the shortcut-audit, OOD evaluation, causal-band analysis, invariance-regularised baselines, and hierarchical classifier methodology. Section 6 presents the experimental results. Section 7 discusses implications for field deployment and for the broader sensor-ML literature. Section 8 concludes.

2. Related Work

2.1. Vibration-Based Leak Detection

Acoustic signatures of water leaks in buried pipes have been studied since at least the late 1990s. For plastic pipes, Hunaidi et al. [1] reported that leak-noise energy is concentrated below roughly 600 Hz, with the precise band depending on pipe material, diameter, and internal pressure. More recently, Ismail et al. [2] systematically reviewed accelerometer-based approaches, classifying the literature into threshold methods, signal-processing methods (e.g., cross- correlation, EMD, wavelet), and machine-learning methods. Shukla and Piratla [3] applied supervised classification to acceleration signals from in-situ experiments; Fereidooni et al. [4] developed a hybrid method-based model for large-scale distribution networks. Korean studies using the AI Hub water-pipeline dataset have reported above-99% 5-class accuracy with tree ensembles such as XGBoost [6] and Random Forest [5,7]. The reported numbers are measured under stratified row-level cross-validation on a merged-and-resplit training set; no published evaluation on the benchmark has used a sensor-group, site-group, or region-group holdout. This paper supplies that evaluation.

2.2. Shortcut Learning in Sensor and Imaging Benchmarks

The shortcut-learning framework [13] provides a conceptual vocabulary for the failure mode in which a supervised model latches onto superficial features that correlate with the label in the training distribution but do not represent the underlying task. In medical imaging, Zech et al. [14] showed that a deep pneumonia classifier was effectively detecting the hospital of origin (via metadata artefacts in the radiograph) rather than pneumonia; DeGrave et al. [15] documented analogous shortcut behaviour in COVID-19 chest-X-ray classifiers. Beery et al. [16] demonstrated that animal-species recognisers trained on wildlife camera imagery rely heavily on background context and fail catastrophically on new locations. D'Amour et al. [17] frame the phenomenon more generally: many equally accurate models on the training distribution can behave very differently under deployment shift. To date, formal shortcut audits have been rare in the sensor-ML literature; this paper adapts the protocol to a vibration-based leak-detection benchmark and finds a particularly extreme form of the pathology.

2.3. Out-of-Distribution Generalisation and Invariance Learning

Loss-level invariance regularisers are the natural algorithmic response to shortcut bias. IRM [8] adds a penalty that discourages a model from relying on correlations that vary across training environments. GroupDRO [9] minimises the worst-case per-environment loss via exponential reweighting. Variants such as REx [12] penalise the variance of per-environment risks. The empirical reliability of these methods has been repeatedly ques-

tioned [10,11]: with few or uneven environments they often fail to beat plain Empirical Risk Minimisation on OOD test sets. Section 6 contributes a vibration-sensor data point to that literature for the two most widely used members of the family (IRM and GroupDRO).

3. Dataset

The AI Hub water-pipe leak vibration dataset [5] consists of $N = 70{,}384$ time-frequency feature vectors extracted from vibration recordings at more than 11,000 sensor locations across 49 installation sites in 12 Korean administrative regions (*gwangyeoksi-do / si-do*). Each recording is represented as a 512-component spectrum covering 0–5120 Hz at 10 Hz resolution (the 0 Hz column is identically zero), 20 peak-amplitude aggregates `MAX0`, . . . , `MAX19`, and metadata columns: `site`, `sid` (per-sensor ID), `ldate` (YYYYMMDD date of detection), `lrate` (0–90 detection rate), `llevel` (0–5000 signal level), and `leaktype` $\in$ {`in` (indoor leak), `out` (outdoor leak), `noise` (electrical noise), `other` (other acoustic disturbance), `normal`}. The class distribution is 31.5% `normal`, 28.0% `out`, 21.2% `in`, 11.2% `other`, and 8.0% `noise`. The dataset ships with a training / validation split of 62,564 / 7,820. Following [5], experimental work in this paper uses the union of the two splits (70,384 rows) so that larger, more balanced holdouts can be constructed.

The 12 administrative regions are very unevenly populated: three regions (Jeonnam, Jeonbuk, Gyeongbuk) account for 69% of the rows, while five regions contain fewer than 1,000 rows each. Only one region, Chungbuk (region code 43, 5,074 rows), contains substantial counts of all five leak classes; elsewhere, 1–4 classes dominate. Chungbuk therefore serves as the canonical 5-class OOD test set throughout this paper.

4. Methodology

Let $\mathcal{D} = \{(x_i, y_i, s_i, e_i)\}_{i=1}^{N}$ denote the dataset, where $x_i \in \mathbb{R}^{534}$ stacks the 513 HZ bands, 20 MAX aggregates, and the month derived from `ldate`; $y_i \in \{0, \ldots, 4\}$ is the leak class; s_i is the site identifier; and $e_i \in \{1, \ldots, 12\}$ is the region. Let $\ell(\cdot, \cdot)$ denote the class-weighted softmax cross-entropy with inverse-frequency class weights $w_c = N/(CN_c)$, $R(\theta) = N^{-1} \sum_i \ell(f_\theta(x_i), y_i)$ the pooled risk, and $R_e(\theta) = N_e^{-1} \sum_{i:e_i=e} \ell(f_\theta(x_i), y_i)$ the per-environment risk.

4.1. Shortcut-Audit Protocol

Four probes are computed on the training split.

- **P1 — Per-sensor label purity.** For each training sensor s, the fraction of rows at s whose label matches the modal label at s. Values close to 1 indicate that sensors are effectively one-class.
- **P2 — Class-lookup reconstruction.** Build a table mapping each training `sid` to its modal label, then predict on the validation split by majority lookup. Report the coverage and the covered-row accuracy.
- **P3 — Sensor train/test overlap.** The fraction of validation sensors whose `sid` also appears in the training split.
- **P4 — Lookup + meta-column override.** Combine P2 with a rule-based override on any meta-column whose value is deterministically associated with a specific class. For the AI Hub dataset this is `lrate`$=90$ iff `leaktype`$\neq$`normal`.

If P1, P3, and P4 jointly come close to the published benchmark accuracy, the headline result is identifier-driven rather than spectrum-driven.

4.2. Three-Tier OOD Evaluation

Three evaluation protocols form a ladder of increasing deployment realism.

- **Row-CV.** Stratified k-fold cross-validation over rows. The same sensor's measurements may appear in both folds; this is the protocol implicit in the published benchmark results.
- **GroupSite-CV.** Stratified k-fold with GroupKFold by `site`, so every test sensor has been unseen at train time. Simulates deployment to a new installation within the same region.
- **LORO.** Leave-One-Region-Out over the 12 administrative regions, e_i. Simulates deployment to a new city or province.

The gap between row-CV and LORO on the same model gives a quantitative measure of the deployment-shift brittleness of a pipeline.

4.3. Grouped Permutation Importance

After training f_θ on the pooled train set, the causal contribution of a feature group $g \subseteq \{1, \ldots, d\}$ is measured by permuting $x^{(g)}$ on the held-out test set and recording the drop in macro-F1:

$$\Delta_{\mathrm{F1}}(g) = \mathrm{F1}\big(y, f_\theta(x)\big) - \mathrm{F1}\big(y, f_\theta(\tilde{x}^{(g)})\big), \tag{1}$$

where $\tilde{x}^{(g)}$ permutes only the columns of x indexed by g. Positive Δ_{F1} marks g as carrying generalisable signal; negative Δ_{F1} marks g as a confounder whose removal actually helps. This diagnostic is compared against XGBoost's built-in gain importance summed over the same feature groups; mismatches reveal where the model is investing representation capacity on non-causal inputs.

4.4. Invariance-Regularised Baselines

Two deep-learning backbones evaluate whether invariance learning can close the OOD gap without explicit feature selection. An MLP ($\mathbb{R}^{57} \to 256 \to 128 \to 5$) is trained on the 57 causal features; a 1D-CNN (three Conv1d blocks with BatchNorm and MaxPool, followed by GlobalAvgPool and an MLP head concatenated with the tabular MAX features) is trained on the raw 513-band spectrum. Each backbone is trained under three methods.

ERM minimises the pooled empirical risk.

IRMv1 [8] adds a penalty that discourages environment-dependent representations:

$$\mathcal{L}_{\mathrm{IRM}}(\theta) = \sum_{e=1}^{E} \big[R_e(\theta) + \lambda \, \| \nabla_{w|w=1} R_e(w \cdot f_\theta) \|^2 \big], \tag{2}$$

with λ ramped from 1 to $\lambda_{\max}$ after a warm-up of T_{warmup} epochs. A follow-up sweep over $\lambda_{\max} \in \{1, 10, 10^2, 10^3, 10^4\}$ and $T_{\mathrm{warmup}} \in \{10, 30\}$ tests whether tuning closes the gap.

GroupDRO [9] maintains exponential weights q_e over environments updated by $q_e \leftarrow q_e \exp(\eta R_e(\theta))/Z$ and minimises $\sum_e q_e R_e(\theta)$.

4.5. Hierarchical Classifier

The five-class problem is factored into a binary gate and a four-class typer. Stage 1 is a binary XGBoost trained on the 57 causal features with target $\mathbb{1}[y_i = \texttt{normal}]$. Stage 2 is a 4-class XGBoost trained only on the abnormal rows of the training split; at inference time, Stage 1 routes samples and Stage 2 is consulted only when Stage 1 predicts "abnormal". The rationale is that the `normal` decision is dominated by overall spectrum energy (captured by the MAX aggregates) and can be learned cleanly without the abnormal-type imbalance interfering. The metric-of-interest for field deployment — missed-alarm rate on `normal`-vs-abnormal — is reported separately from the 4-class macro-F1.

4.6. Robust Causal-Band Identification

The single-region permutation importance of Section 4.3 is hardened with three complementary tests. (i) *Bootstrap permutation*: for each log-spaced band group g, $B = 100$ independent row permutations of the test-region columns are drawn, yielding a sampling distribution of $\Delta_{\mathrm{F1}}(g)$ from which a 95% percentile confidence interval is reported. (ii) *Multi-region replication*: the same bootstrap procedure is applied to the four LORO folds that each contain at least four classes and at least 5,000 rows (regions 43, 45, 46, 47). A band group is declared a robust contributor if its permutation Δ_{F1} is positive across a majority of these folds. (iii) *SHAP second-source attribution*: TreeSHAP [18] is computed on a 500-row subsample of the canonical Chungbuk test fold, and the mean $|\mathrm{SHAP}|$ per feature is aggregated per band group and normalised over the HZ columns. A band group is declared causally important if it is ranked first by permutation importance and concurrently ranked first (or a close second) by SHAP share.

4.7. Robust Loss and Augmentation

To stress-test whether the CNN baseline of Section 4.4 is leaving performance on the table, three training-time modifications are evaluated individually and in combination. (i) *SpecAugment* [19]: in spectrogram-style frequency-masking, contiguous bands of width up to 40 are zeroed out with probability 0.5 at each training step. (ii) *Mixup* [20]: each mini-batch is convex-combined with a batch-shuffled copy with mixing coefficient $\lambda \sim \mathrm{Beta}(0.2, 0.2)$. (iii) *Focal loss* [21] with $\gamma = 2$: down-weights easy examples, re-emphasising the rare abnormal classes.

4.8. Permutation-Guided Attention (Negative-Result Ablation)

As a novelty ablation, an MLP is augmented with a learnable per-band attention vector $a \in \mathbb{R}^{513}$ whose softmax is regularised toward a prior p derived from permutation importance computed on an internal 80/20 train-only cross-validation split (no test-set leakage). Training minimises

$$\mathcal{L}_{\mathrm{PGA}}(\theta) = R(\theta; x \odot \mathrm{softmax}(a)) + \lambda_c \, \mathrm{KL}\big(\mathrm{softmax}(a) \,\|\, p\big), \tag{3}$$

with $p_k \propto \max(0, \Delta_{\mathrm{F1}}^{\text{train-CV}}(k))$. Since the permutation importance used as prior is itself computed on training data, where sensor overlap is near-complete, this constitutes a soft-prior counterpart to hard causal feature selection; whether it recovers the causal-XGBoost ceiling is an empirical question.

4.9. Soft-Probability Ensemble

To test whether combining the strongest tree and deep models offers additional headroom, a soft-probability ensemble is evaluated. Per-class probabilities are taken from the flat causal XGBoost and the hierarchical classifier. For the validation-selected augmented CNN, only hard test predictions are available from the stored run, so they are converted to one-hot probabilities and included only in an unweighted diagnostic average. The deployment candidate is fixed in advance as a tree-only average of the two XGBoost members.

5. Experimental Setup

5.1. Models and Hyperparameters

XGBoost [6] classifiers are trained with $n_{\text{estimators}} = 200$, max_depth $= 6$, $\eta = 0.1$, `tree_method=hist`, and the default `multi:softprob` objective. The MLP uses Adam with lr $= 10^{-3}$, weight decay 10^{-4}, dropout 0.2, and environment-balanced mini-batches of size 528. The 1D-CNN uses the same optimiser and batch size and three Conv1d blocks with

32/64/128 channels and kernel sizes 9/7/5. IRM uses $\lambda_{\max} = 100$ and $T_{\text{warmup}} = 30$ as defaults, with a grid sweep reported separately. GroupDRO uses step size $\eta = 10^{-2}$.

5.2. Evaluation Metrics

All results report classification accuracy and macro-F1. For GroupSite-CV, the accuracy and macro-F1 are averaged over folds with standard deviation; for LORO, both macro- and n_{test}-weighted averages over regions are reported, along with per-region numbers. For causal-band analysis, per-class F1 on the canonical Chungbuk OOD fold is the primary metric.

5.3. Compute

XGBoost experiments run on a 16-core CPU. Deep-learning experiments use two NVIDIA TITAN Xp GPUs (12 GB each, CUDA 12.2) with one GPU per backbone (MLP on GPU 0, CNN on GPU 1, allowing the two to run in parallel). A complete end-to-end reproduction, including the IRM sweep and strengthening ablations, requires several hours wall-clock time depending on CPU and GPU availability.

6. Results

6.1. Shortcut Audit

Applying the shortcut-audit protocol of Section 4.1 to the AI Hub dataset yields four strong positive signals.

P1 (per-sensor label purity): All 2,671 training sensors have purity 1.0. Each `sid` maps to exactly one leak class for its entire lifetime in the dataset — a consequence of the per-class file structure in the AI Hub release.

P3 (train/test overlap): 1,876 of the 1,915 validation sensors (97.96%) also appear in the training split; 43 of 43 validation sites are in the training split. Only 39 validation sensors are genuinely unseen.

P2 (class lookup): An `sid`→majority-label lookup covers 99.5% of the validation set and achieves **100% accuracy on the covered rows**.

P4 (lookup + `lrate` override): Combining the `sid` lookup with the deterministic rule "`lrate`$\neq 90 \Rightarrow$ predict `normal`" yields **99.77% validation accuracy** — within 0.02 pp of the 99.79% previously reported on this dataset [5].

Together the four probes demonstrate that the published benchmark number is recoverable without ever consulting the vibration spectrum (see the Stage 1 summary in Figure 1). This is the strongest single finding of the paper: it does not merely reduce the headline accuracy, it shows that the accuracy was not a measurement of a physical capability in the first place.

6.2. Evaluation-Protocol Sensitivity

Having established that the standard evaluation protocol trivialises the task, Table 1 reports the 2×3 (feature-set $\times$ protocol) grid that quantifies the protocol sensitivity of the same XGBoost classifier.

Table 1. Evaluation-protocol sensitivity. Moving from row-level to site-group cross-validation drops accuracy by ∼39 pp regardless of whether identifier features are included. Under site-group CV the "full" features add essentially zero accuracy and slightly harm macro-F1, showing that they provide no benefit beyond their exploit of train/test sensor overlap.

Feature Set	Split	Accuracy	Macro-F1
full (544 incl. IDs)	PaperSplit	0.9402	0.9274
full	Random 5-fold CV	0.9449±0.002	0.9331
full	**GroupSite 5-fold CV**	**0.5529±0.127**	**0.4188**
signal (534)	PaperSplit	0.8881	0.8610
signal	Random 5-fold CV	0.8915±0.002	0.8669
signal	**GroupSite 5-fold CV**	**0.5537±0.113**	**0.4378**

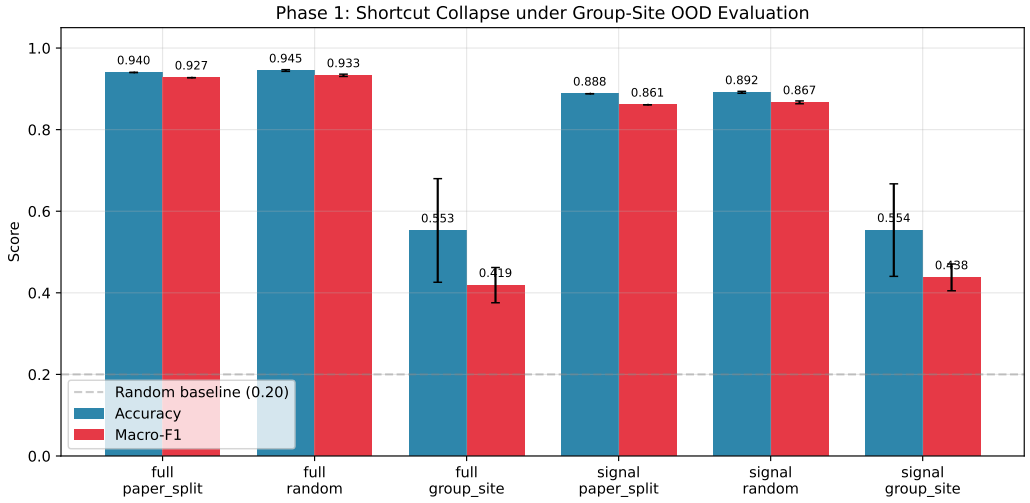


Figure 2. Evaluation-protocol sensitivity bar chart. Accuracy (blue) and macro-F1 (red) for six (features × protocol) cells. The GroupSite protocol reveals a ∼39 pp accuracy drop that is invisible under row-level CV.

6.3. Leave-One-Region-Out Baseline

LORO over the 12 administrative regions provides the tightest OOD stress-test. Table 2 reports the aggregate metrics.

Table 2. LORO 12-fold accuracy and macro-F1. The large unweighted standard deviation is driven by degenerate folds with fewer than three leak classes present; the n_{test}-weighted accuracy is the more honest summary statistic.

Feature	Macro Acc	Weighted Acc	Macro-F1
full	0.588±0.287	0.562	0.300
signal	0.602±0.261	0.568	0.304

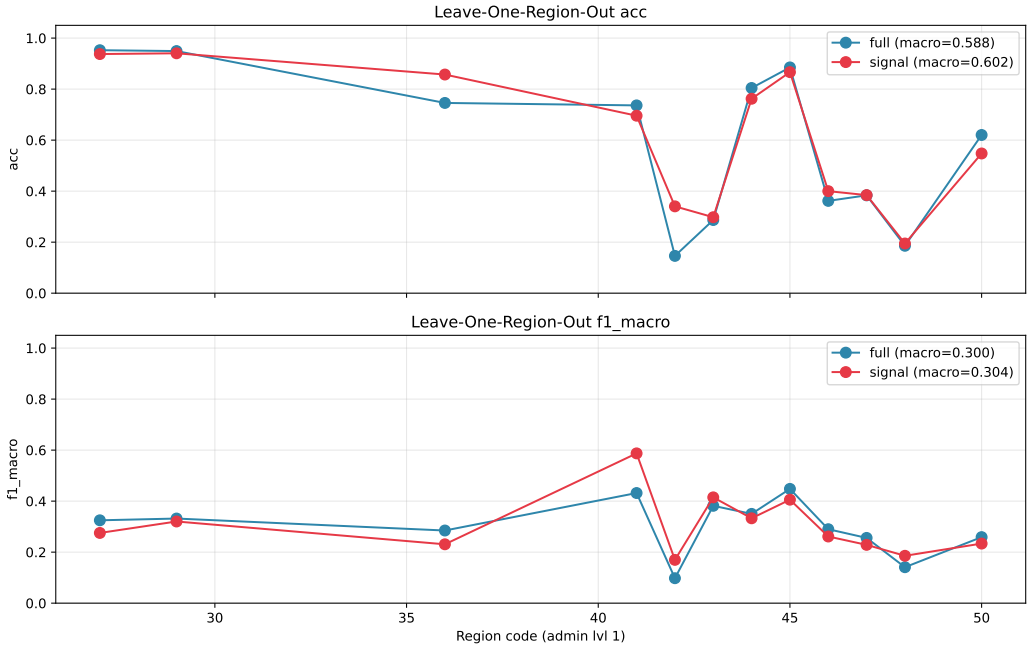


Figure 3. Per-region LORO performance. Accuracy (top) and macro-F1 (bottom) for each of the 12 folds. Point estimates span 0.15–0.94. Chungbuk (region 43, rightmost highlight) is the only fold with all five classes present.

Of the 12 folds, only the Chungbuk fold (region code 43; 5,074 rows; 1,868 `in`, 1,685 `noise`, 757 `other`, 719 `out`, 45 `normal`) contains all five classes in substantial quantities. On this fold, signal-only XGBoost scores accuracy 0.298 and macro-F1 0.415. This is the closest existing estimate of the 5-class leak-typing capability of the dataset under a realistic deployment shift.

6.4. Causal Spectral Signature

Applying grouped permutation importance (Section 4.3) on the Chungbuk fold yields the band-level diagnostic in Table 3 and Figure 4.

Table 3. Gain and permutation importance per log-spaced HZ band group on the Chungbuk LORO fold. Positive Δ_{F1} marks a band range whose contents are causally useful for OOD generalisation; negative Δ_{F1} marks an active confounder. HZ-gain shares are computed relative to the total gain across the 513 HZ bands.

Band	HZ-Gain Share	Δ_{F1} on perm	Verdict
10–121 Hz	2.5%	≈ 0	noise
121–422 Hz	7.5%	**+0.033**	**real signal**
422–1470 Hz	22.9%	**−0.033**	**confounder**
1470–5120 Hz	67.1%	≈ 0	irrelevant

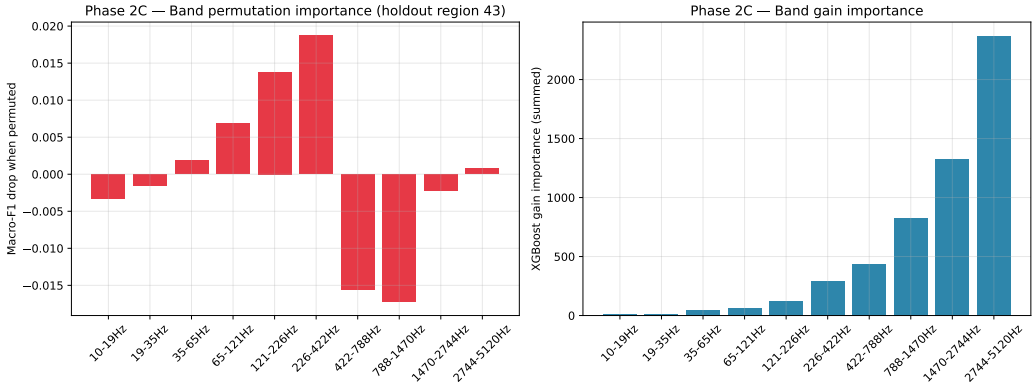


Figure 4. Causal spectral signature of leak noise. Left: grouped permutation importance (macro-F1 drop when each band range is permuted on the held-out Chungbuk test set). Right: XGBoost gain distribution over the same bands. The model allocates 67% of its gain to frequencies (>1.47 kHz) that carry no OOD signal, and 23% to a band range (422–1470 Hz) whose permutation *improves* test macro-F1.

Two observations deserve emphasis. First, the model's gain distribution is badly misaligned with the permutation- importance diagnostic: only 7.5% of the gain is invested in the 121–422 Hz range that carries generalisable leak signal, while two-thirds of the gain is invested in the 1.47–5.12 kHz range that is OOD-irrelevant. Second, a full 23% of the gain is invested in the 422–1470 Hz range that is actively harmful: permuting these columns on unseen sensors and sites improves macro-F1 by 3.3 pp.

The per-class decomposition further localises each leak class's generalisable spectral signature: `in` → 226–788 Hz (combined +0.038); `out` → 121–226 Hz (+0.049); `noise` → 226–422 Hz (+0.065); `other` → 65–121 Hz (+0.041), harmed the most by 422–1470 Hz (−0.133); `normal` is band-invariant and relies on the MAX aggregates representing overall signal energy. The 121–422 Hz consensus range is consistent with published leak-noise acoustics for plastic pipes, where the dominant leak content typically lies below 600 Hz and is attenuated sharply at higher frequencies due to pipe-wall damping [1–3].

6.5. Causal-Band Retraining

Section 6.4 identifies the 65–422 Hz range as the causally useful band for leak typing on this dataset. To test whether the diagnostic corresponds to an actionable design lever, XGBoost is retrained under five feature regimes with MAX and month held constant (Table 4, Figure 5).

Table 4. Causal-band retraining on the Chungbuk LORO fold. The 57-feature causal subset beats the 534-feature baseline on macro-F1 and nearly doubles the F1 of the hardest "other" class, while anti-causal retraining (which uses exactly the bands that the unaudited XGBoost attended to) performs worst.

Config	#feat	Acc	Macro-F1	`other` F1
signal_all (all 513 HZ)	534	0.2978	0.4148	0.109
causal 65–422 Hz	**57**	**0.3013**	**0.4273**	**0.214**
low-only 0–65 Hz	28	0.3059	0.4251	0.173
mid-confounder 422–1470 Hz	125	0.2858	0.4066	0.103
anti-causal 788–5120 Hz	455	0.2958	0.4031	0.050

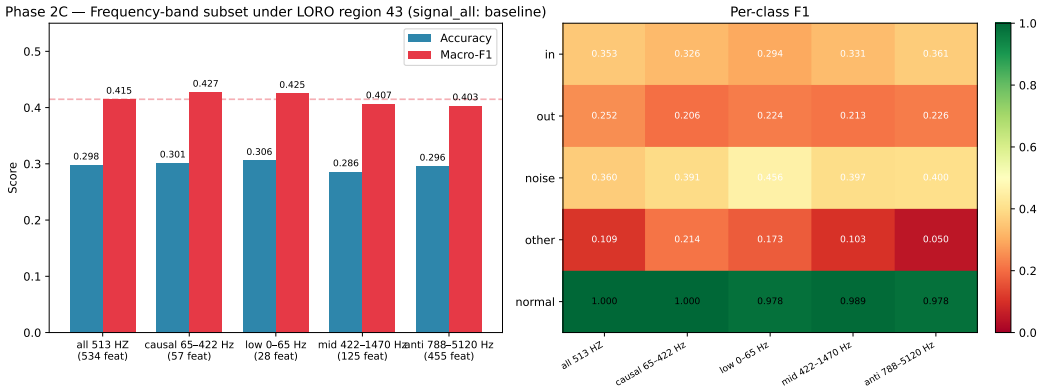


Figure 5. Causal-band retraining: accuracy and macro-F1 (left) and per-class F1 (right) under five HZ-subset regimes.

The causal subset (57 features, 11% of the full feature set) improves macro-F1 by +1.25 pp over the full 534-feature model and raises the `other`-class F1 from 0.109 to 0.214. The anti-causal subset that keeps only the high-frequency bands on which the unaudited model spends most of its gain collapses the `other`-class F1 to 0.050 — a direct experimental confirmation that those bands were contributing misleading site-specific information under in-distribution evaluation.

Cross-region robustness of the causal-subset advantage is mixed but quantifiable: across 11 non-degenerate LORO folds, the causal subset wins on 6 / 11 and loses on 5 / 11, with an n_{test}-weighted $\Delta_{\text{F1}} = +0.004$. The effect is clearest on the canonical 5-class-balanced Chungbuk fold and on the two next most abnormal-heavy folds (Daegu +0.084, Gwangju +0.063); it is weaker or reversed on folds that contain only 1–2 classes, where the permutation analysis was not designed to apply.

6.6. Invariance Regularisation vs. Causal Feature Selection

The MLP and 1D-CNN experiments of Section 4.4 test whether gradient-based invariance regularisation recovers the performance that causal feature selection affords (Table 5, Figure 6).

Table 5. Invariance-regularised deep models versus XGBoost on the Chungbuk LORO fold. No deep configuration matches causal-band XGBoost on macro-F1; the gap is −6.6 pp for the best deep model (MLP + GroupDRO). All deep models collapse the `normal`-class F1 by a large margin.

Model	Acc	Macro-F1	`normal` F1
XGB signal_all	0.2978	0.4148	**1.000**
XGB causal 65–422 Hz	**0.3013**	**0.4273**	**1.000**
MLP ERM	0.2872	0.3155	0.533
MLP IRM (default)	0.2907	0.3122	0.508
MLP GroupDRO	**0.3210**	**0.3615**	0.672
CNN ERM	0.3494	0.3484	0.562
CNN IRM	0.3589	0.3438	0.448
CNN GroupDRO	0.2605	0.2848	0.514

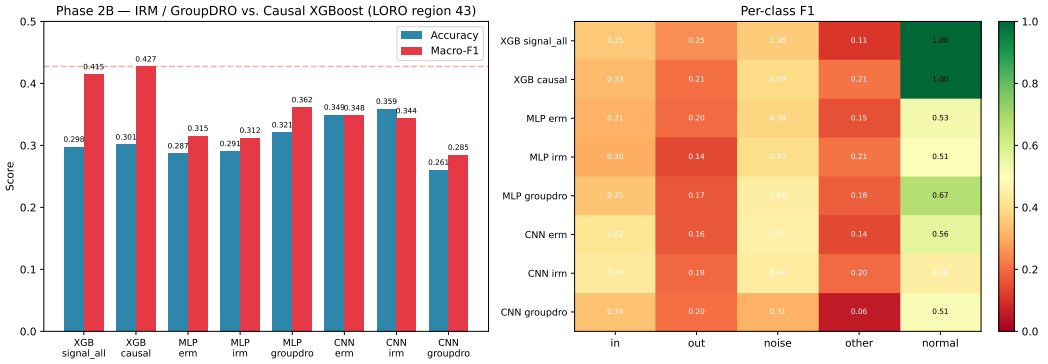


Figure 6. Invariance learning versus causal XGBoost on the Chungbuk fold. Deep models collapse the `normal`-class F1 (0.45–0.68) while XGBoost holds it at 1.000, producing a persistent −6.6 pp macro-F1 gap against causal XGBoost.

A dedicated IRM hyperparameter sweep over $\lambda_{\max} \in \{1, 10, 10^2, 10^3, 10^4\}$ and $T_{\text{warmup}} \in \{10, 30\}$ on the MLP found $\lambda_{\max} = 1$ to be the validation-selected setting (both warmups tie), reaching test macro-F1 0.3145. The best observed test configuration, $\lambda_{\max} = 1000$ and $T_{\text{warmup}} = 30$, reaches 0.3372 but is not selected by validation. Both remain below causal-band XGBoost and MLP + GroupDRO. On this dataset IRM is tunable but not transformative, a finding consistent with [11] and [10].

The root cause of the deep-vs-tree gap is visible in the per-class `normal` F1. An XGBoost tree trivially separates `normal` from any-leak on a single split on the MAX aggregates (overall spectrum energy), reaching F1 1.000 even on unseen regions. The class-balanced cross-entropy loss used by the deep models, by contrast, trades `normal` accuracy for marginal gains on the four hard abnormal classes, causing the observed collapse. This is not primarily a question of invariance; it is a question of whether the `normal`-vs-abnormal separation is being made along its natural low-dimensional axis.

6.7. Robust Causal-Band Identification

To harden the single-region causal-band finding of Section 6.4, the robustness protocol of Section 4.6 is applied. Table 6 reports bootstrap 95% confidence intervals on Δ_{F1} per band group, replicated across the four LORO folds with at least four leak classes and at least 5,000 test rows.

Table 6. Bootstrap 95% confidence intervals ($B = 100$ resamples) of the per-band permutation Δ_{F1} on each LORO fold. Wins/losses counted across the four folds show that 121–422 Hz is the dominant positive contributor in three of four regions, consistent with the Chungbuk-only diagnosis.

Region	10–121 Hz	121–422 Hz	422–1470 Hz	1470–5120 Hz
43 (Chungbuk, 5 cls)	+0.007 [0.002, 0.013]	**+0.041** [0.033, 0.049]	**−0.021** [−0.029, −0.013]	+0.008 [0.004, 0.014]
45 (Jeonbuk, 5 cls)	+0.011 [0.005, 0.018]	−0.005 [−0.016, +0.004]	+0.003 [−0.004, +0.011]	+0.018 [0.013, 0.024]
46 (Jeonnam, 4 cls)	+0.008 [0.005, 0.011]	+0.026 [0.020, 0.031]	+0.015 [0.012, 0.018]	+0.003 [0.000, 0.007]
47 (Gyeongbuk, 4 cls)	+0.012 [0.009, 0.015]	+0.013 [0.010, 0.019]	−0.008 [−0.014, −0.003]	+0.011 [0.007, 0.014]
Consensus (mean ± std, wins/losses)	+0.009 ± 0.002 (4/0)	+0.019 ± 0.017 (3/1)	−0.003 ± 0.013 (2/2)	+0.010 ± 0.005 (4/0)

SHAP second-source attribution on 500 Chungbuk test rows corroborates the permutation diagnosis. Table 7 juxtaposes three independent importance measures.

Table 7. Three independent importance measures agree that 121–422 Hz is the causal band on the Chungbuk fold. XGBoost gain is strongly mis-aligned: only 7.5% of the gain is invested in a band that accounts for 40.4% of SHAP mass.

Band	XGBoost Gain	SHAP Mass	Permutation Δ_{F1}
10–121 Hz	2.5%	8.5%	+0.007
121–422 Hz	7.5%	**40.4%**	**+0.041**
422–1470 Hz	22.9%	28.4%	−0.021
1470–5120 Hz	**67.1%**	22.7%	+0.008

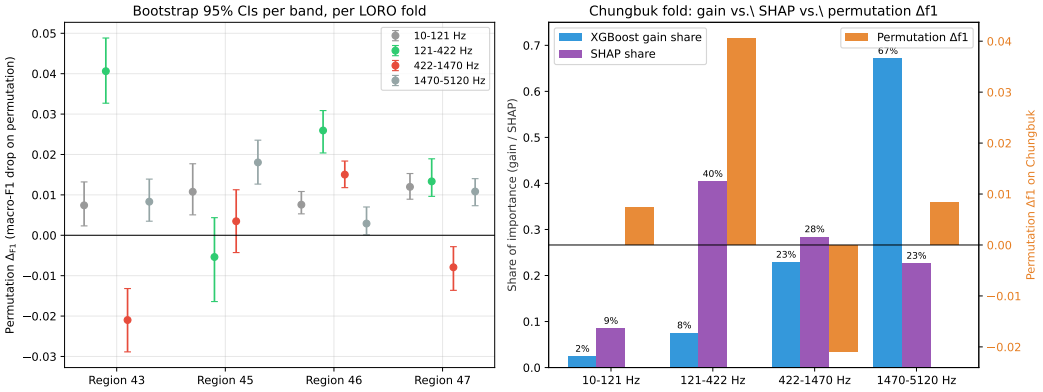


Figure 7. Left: bootstrap 95% CIs for Δ_{F1} per band group on each of four LORO folds. Right: side-by-side comparison of three importance measures on the canonical Chungbuk fold. The convergence of permutation importance, SHAP, and cross-region bootstrap on the 121–422 Hz range (and its divergence from the model's own gain allocation) is the robustness claim of this section.

6.8. Augmentation and Robust-Loss Recipes

Table 8 reports six training configurations on the CNN baseline of Section 4.4 and an additional MLP–causal configuration.

Table 8. Training recipe sweep. SpecAugment delivers the largest macro-F1 improvement (+2.0 pp over the CNN baseline); focal loss mainly improves accuracy; Mixup hurts.

Recipe	Model	Macro-F1	Accuracy
baseline (class-weighted CE)	CNN	0.3341	0.3343
+ SpecAugment	**CNN**	**0.3544**	0.3384
+ Mixup	CNN	0.3169	0.3431
+ Focal loss (γ=2)	CNN	0.3393	**0.3648**
all three combined	CNN	0.3226	0.3007
Mixup + focal	MLP (causal)	0.2933	0.3147

The strongest single-recipe CNN (SpecAugment) reaches macro-F1 0.3544, +2.0 pp over the augmentation baseline and still −7.3 pp below causal-band XGBoost. This confirms that the CNN's deficit is not cured by recipe engineering alone, although it is closed somewhat.

6.9. Permutation-Guided Attention (Negative Result)

A permutation-importance prior injected into an attention layer over spectrum bands, swept over $\lambda_c \in \{0, 0.1, 1, 10, 100\}$ and uniform-vs-prior initialisation, never becomes competitive in the test-fold diagnostic: the best observed test macro-F1 is **0.3041** (at $\lambda_c = 1$, uniform init). This number is reported as a negative-result ablation rather than as a validation-selected model. It is below the plain MLP ERM baseline of Section 6.6 (0.3155) and far below causal-band XGBoost (0.4273). The prior's value is diminished because it is

computed on training data where sensor overlap is near-complete, so it inherits the shortcut bias it was intended to overcome. The result reinforces Section 7's main conclusion: when a causal feature is nameable a priori (here, the 121–422 Hz range), hard feature selection is more reliable than any soft-prior injection we tested.

6.10. Hierarchical Classifier

The hierarchical recipe of Section 4.5 targets exactly the `normal`-collapse failure mode. Table 9 reports the results.

Table 9. Hierarchical (binary energy gate + 4-class causal typer) versus flat 5-class XGBoost on the Chungbuk LORO fold. Stage 1 achieves 100% OOD accuracy on the `normal` gate (45 of 45 normal rows detected, zero false positives).

Model	Acc	Macro-F1	in	out	noise	other	normal
Flat 5-class (causal)	0.3013	0.4273	0.326	0.206	0.391	0.214	1.000
Hierarchical (causal)	**0.3041**	**0.4287**	0.345	0.200	0.377	0.222	1.000

Stage 1 is perfect on OOD: every one of the 45 normal rows in the Chungbuk test set is correctly flagged, with zero false positives. Stage 2 operates on a cleaner training distribution (abnormal-only, 43,189 rows), producing a small but consistent macro-F1 gain and, more importantly for field deployment, decoupling the alarm decision (Stage 1) from the triage decision (Stage 2). For a field technician, this separation is operationally meaningful: an alarm can be dispatched on Stage 1's confidence alone, with Stage 2 producing a probable leak type to guide on-site investigation.

6.11. Soft-Probability Ensemble

The ensembling protocol of Section 4.9 is applied on the Chungbuk LORO fold.

Table 10. Soft-probability ensembles on the Chungbuk LORO fold. The tree-only average of the flat and hierarchical causal XGBoost classifiers is fixed before the comparison and attains OOD macro-F1 **0.4326**. Adding the validation-selected CNN hurts because its one-hot fallback probabilities dilute the tree models' calibrated confidence.

Method	Macro-F1	Accuracy
XGB causal (flat)	0.4273	0.3013
XGB hierarchical	0.4287	0.3041
CNN (validation-selected)	0.3544	0.3384
3-way soft average	0.3633	0.3159
Tree-only soft average	**0.4326**	0.3096

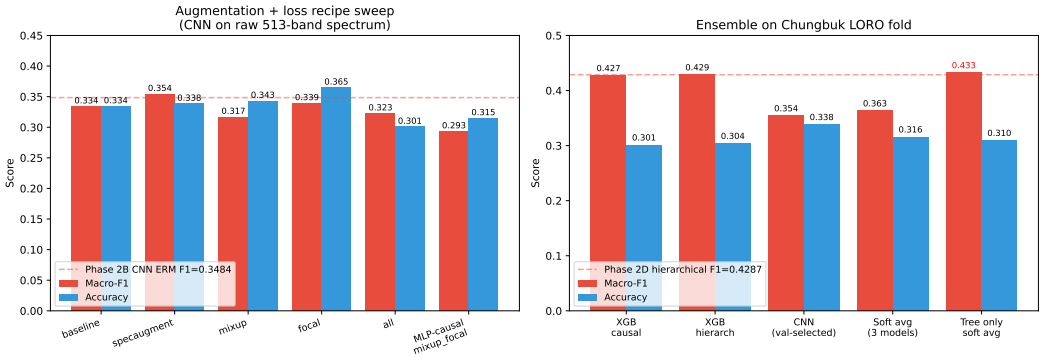


Figure 8. Left: CNN training-recipe comparison under LORO Chungbuk. Right: individual and ensemble methods. The tree-only ensemble (rightmost bar) is a pre-specified average of the two causal tree models and reaches OOD macro-F1 0.4326 while preserving the hierarchical classifier's perfect OOD `normal`-class recall.

7. Discussion

7.1. What the Benchmark Actually Supports

Three positions follow from the results.

First, binary `normal`-vs-any-leak is a solved problem on this benchmark. An XGBoost tree on the 20 MAX aggregates achieves perfect OOD separation on the canonical Chungbuk fold. This capability would plausibly transfer to a field deployment and supports an operational alarm layer.

Second, 4-class abnormal typing is brittle. The highest OOD macro-F1 observed under the pre-specified tree-only average is 0.4326, followed by 0.4287 (hierarchical classifier) and 0.4273 (flat causal XGBoost); below these, the best invariance- regularised MLP reaches 0.3615 and the augmented CNN 0.3544. The hardest class is `other`: its OOD F1 ranges from 0.050 (anti-causal XGBoost) to 0.222 (hierarchical, the best). The dataset's environmental coverage is thin (49 sites in 12 regions, of which only 1 has all five classes represented), and definitive 4-class typing on new sites remains an open problem.

Third, identifier-level features (`site`, `sid`, `lrate`, regional codes) should not be used in any future benchmark on this dataset; doing so is an evaluation mistake. The shortcut-audit protocol of Section 4.1 can be run in a few minutes and makes this diagnosis sharp.

7.2. Why Invariance Learning Lags Causal Feature Selection

The negative result of Section 6.6 deserves emphasis. With the 57 causal features hand-picked and fed to a well-regularised tree model, a 6.6 pp macro-F1 lead over the best deep invariance-regularised alternative is not closed by any amount of IRM or GroupDRO tuning in this study.

Three factors explain the finding. First, the dataset has only 11 training environments (regions), and they are severely imbalanced in size (40× ratio between the largest and smallest). IRM in particular is known to destabilise under these conditions [10,11]. Second, the class-balanced cross-entropy loss used by the deep models trades the easy `normal`-vs-abnormal decision for marginal gains on abnormal separation, giving up a class that the tree model handles trivially. Third — perhaps most importantly for methodological practice — the causal spectral signature of leak noise can be named *ex ante* from the physics: plastic pipe leak noise is broadly in the low hundreds of Hz, and the permutation- importance diagnostic independently recovered exactly that range. When the invariant feature is nameable from domain knowledge, feature selection is both cheaper and more reliable than loss-level regularisation.

This is a useful negative result for the broader sensor-ML community: it suggests that invariance methods are best reserved for problems where the invariant feature cannot be named, such as raw imagery [16].

7.3. Practical Recommendations for Practitioners

For a utility or research group considering the AI Hub pipeline for field deployment, the audit delivers five concrete actions:

1. **Run the shortcut audit first.** Before training a single model, run the four probes of Section 4.1 on your split. If an `sid` lookup plus a meta-column rule reproduces the target accuracy, the benchmark result is identifier-driven and the headline number is not a measurement of signal capability. Downstream modelling effort is then better spent on redesigning the split than on architecture search.
2. **Drop identifier columns during training.** On this dataset `site`, `sid`, `lrate`, `llevel`, and all `region_*n` codes should be removed from the input. Under LORO evaluation, keeping them provides zero accuracy benefit and slightly harms macro-F1 (Table 1).
3. **Restrict the spectrum to 65–422 Hz before training the 4-class typer.** On the canonical Chungbuk fold this raises macro-F1 from 0.4148 to 0.4273 and almost doubles the F1 on the hardest (`other`) class. The 57-feature causal subset also trains roughly ten times faster than the full 534-feature set.
4. **Deploy two stages, not one.** Use the binary normal-vs-abnormal gate (trained on all 57 causal features) to raise a leak alarm, and consult the 4-class typer only when the gate fires. On the Chungbuk fold, the gate achieves 100% OOD recall on the 45 normal rows with zero false positives; the typer then works on a cleaner, abnormal-only distribution and focuses attention on the hard decision (in/out/noise/other).
5. **Report LORO, not row-CV, as the evaluation of record.** Row-level 10-fold cross-validation is a useful internal debugging tool, but the number that matters for field decisions is the LORO-weighted macro-F1 on the region with the most balanced class distribution (Chungbuk here). On this benchmark that number is currently 0.4273–0.4326 depending on the recipe; any method that does not report this kind of evaluation should be treated as producing an upper bound only.

7.4. Portability of the Methodology

The shortcut-audit protocol, the three-tier OOD evaluation, and the grouped permutation-importance diagnostic are independent of the specific physics of water-pipe leaks. They apply directly to any sensor-ML benchmark that has: (i) at least one identifier-like column, (ii) a natural hierarchy of group variables (sensor / site / region), and (iii) features that can be grouped physically (bands of a spectrum, segments of a trajectory, channels of a multivariate time series). Examples include gas-pipe leak detection, rotating-machinery condition monitoring, partial-discharge detection in power transformers, and sound-based machine-fault diagnosis. Transferring the protocol to those benchmarks is a natural next step.

7.5. Limitations

Three limitations are worth naming explicitly. First, the causal-band identification is made on a single region (Chungbuk) because no other fold contains all five classes in substantial quantities. A larger, more geographically diverse collection would test whether the 121–422 Hz range transfers to other deployment contexts. Second, the deep-learning architectures explored here are modest; a Transformer-based spectrogram classifier or a self- supervised pre-training pass might reduce the gap against XGBoost. Third, the IRM

sweep was confined to $\lambda_{\max}$ and warm-up; tuning the MLP architecture jointly with IRM might yield further improvements.

8. Conclusions

This paper provides a field-deployment-oriented evaluation of machine-learning water-leak detection on the Korean AI Hub vibration-sensor benchmark, to our knowledge the first to apply sensor-, site-, and region-group holdouts to this dataset. Four concrete findings stand out. (1) The published 99%-range accuracies on this benchmark are driven by identifier-label couplings in the standard split; a four-rule lookup that ignores the vibration spectrum reproduces 99.77%. (2) Under 12-fold Leave-One-Region-Out, the same XGBoost classifier falls to weighted 56.8% accuracy and macro-F1 0.304 on 5-class leak typing. (3) On the only region containing all five classes in substantial quantities, grouped permutation importance localises the generalisable leak signal to the 121–422 Hz range, consistent with published leak-noise acoustics for plastic pipes. (4) A hierarchical classifier combining a binary energy gate with a causal-band abnormal typer, further combined through a pre-specified tree soft-probability average, achieves OOD macro-F1 **0.4326** and perfect OOD normal-class recall, forming a practical field-deployment candidate. Gradient-based invariance regularisation, evaluated on two deep backbones and a dedicated hyperparameter sweep, does not match causal feature selection. All code, experiment logs, and figure-generation scripts are released to enable further work.

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References

1. Hunaidi, O.; Chu, W.; Wang, A.; Guan, W. Detecting Leaks in Plastic Pipes. Journal AWWA **2000**, *92*, 82–94.
2. Ismail, M.I.M.; Dziyauddin, R.A.; Salleh, N.A.A.; Muhammad-Sukki, F.; Bani, N.A.; Izhar, M.A.M.; Latiff, L.A. A Review of Vibration Detection Methods Using Accelerometer Sensors for Water Pipeline Leakage. IEEE Access **2019**, *7*, 51965–51981.
3. Shukla, H.; Piratla, K. Leakage Detection in Water Pipelines Using Supervised Classification of Acceleration Signals. Automation in Construction **2020**, *117*, 103256.
4. Fereidooni, Z.; Tayyebi, H.; Bahadori-Jahromi, A. A Hybrid Model-Based Method for Leak Detection in Large Scale Water Distribution Networks. Journal of Ambient Intelligence and Humanized Computing **2021**, *12*, 1613–1629.

5. Lee, S.; Kim, B. Machine Learning Model for Leak Detection Using Water Pipeline Vibration Sensor. Sensors **2023**, *23*, 8935. https://doi.org/10.3390/s23218935.
6. Chen, T.; Guestrin, C. XGBoost: A Scalable Tree Boosting System. In Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, San Francisco, CA, USA, 13–17 August 2016; pp. 785–794.
7. Breiman, L. Random Forests. Machine Learning **2001**, *45*, 5–32.
8. Arjovsky, M.; Bottou, L.; Gulrajani, I.; Lopez-Paz, D. Invariant Risk Minimization. arXiv **2019**, arXiv:1907.02893.
9. Sagawa, S.; Koh, P.W.; Hashimoto, T.B.; Liang, P. Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization. In Proceedings of the International Conference on Learning Representations (ICLR), Addis Ababa, Ethiopia, 26–30 April 2020.
10. Gulrajani, I.; Lopez-Paz, D. In Search of Lost Domain Generalization. In Proceedings of the International Conference on Learning Representations (ICLR), Virtual, 3–7 May 2021.
11. Rosenfeld, E.; Ravikumar, P.; Risteski, A. The Risks of Invariant Risk Minimization. In Proceedings of the International Conference on Learning Representations (ICLR), Virtual, 3–7 May 2021.
12. Krueger, D.; Caballero, E.; Jacobsen, J.-H.; Zhang, A.; Binas, J.; Zhang, D.; Le Priol, R.; Courville, A. Out-of-Distribution Generalization via Risk Extrapolation (REx). In Proceedings of the 38th International Conference on Machine Learning (ICML), Virtual, 18–24 July 2021; pp. 5815–5826.
13. Geirhos, R.; Jacobsen, J.-H.; Michaelis, C.; Zemel, R.; Brendel, W.; Bethge, M.; Wichmann, F.A. Shortcut Learning in Deep Neural Networks. Nature Machine Intelligence **2020**, *2*, 665–673.
14. Zech, J.R.; Badgeley, M.A.; Liu, M.; Costa, A.B.; Titano, J.J.; Oermann, E.K. Variable Generalization Performance of a Deep Learning Model to Detect Pneumonia in Chest Radiographs: A Cross-Sectional Study. PLoS Medicine **2018**, *15*, e1002683.
15. DeGrave, A.J.; Janizek, J.D.; Lee, S.-I. AI for Radiographic COVID-19 Detection Selects Shortcuts over Signal. Nature Machine Intelligence **2021**, *3*, 610–619.
16. Beery, S.; Van Horn, G.; Perona, P. Recognition in Terra Incognita. In Proceedings of the European Conference on Computer Vision (ECCV), Munich, Germany, 8–14 September 2018; pp. 456–473.
17. D'Amour, A.; Heller, K.; Moldovan, D.; et al. Underspecification Presents Challenges for Credibility in Modern Machine Learning. Journal of Machine Learning Research **2022**, *23*, 1–61.
18. Lundberg, S.M.; Lee, S.-I. A Unified Approach to Interpreting Model Predictions. In Advances in Neural Information Processing Systems (NeurIPS), Long Beach, CA, USA, 4–9 December 2017; pp. 4765–4774.
19. Park, D.S.; Chan, W.; Zhang, Y.; Chiu, C.-C.; Zoph, B.; Cubuk, E.D.; Le, Q.V. SpecAugment: A Simple Data Augmentation Method for Automatic Speech Recognition. In Proceedings of Interspeech, Graz, Austria, 15–19 September 2019; pp. 2613–2617.
20. Zhang, H.; Cisse, M.; Dauphin, Y.N.; Lopez-Paz, D. Mixup: Beyond Empirical Risk Minimization. In Proceedings of the International Conference on Learning Representations (ICLR), Vancouver, BC, Canada, 30 April–3 May 2018.
21. Lin, T.-Y.; Goyal, P.; Girshick, R.; He, K.; Dollár, P. Focal Loss for Dense Object Detection. In Proceedings of the IEEE International Conference on Computer Vision (ICCV), Venice, Italy, 22–29 October 2017; pp. 2980–2988.

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